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The Editors
Ocean Engineering
Elsevier

Dear Editors,

We submit for your consideration the manuscript entitled **"WaveSplat: Grid-Free Spectral Wave Field Estimation from Sparse Shipboard LiDAR"** for publication in *Ocean Engineering*.

Scope and motivation. Real-time wave field estimation from shipboard sensors is a critical enabler for autonomous maritime operations, including proactive motion compensation, route optimisation, and wave-preview feedforward control. Existing methods rely on spatial grid discretisation combined with ensemble Kalman filtering, which introduces interpolation artefacts and fixed spatial resolution. This submission addresses these limitations with a fundamentally different representation.

Principal contributions. WaveSplat represents the ocean wave field as a superposition of spectral components — plane waves constrained by the linear dispersion relation — optimised directly on sparse, irregularly distributed LiDAR observations without any spatial grid. Four novel contributions are presented:

1. A *grid-free spectral estimation framework* with persistent cross-frame tracking via dispersion-based phase prediction, enabling real-time operation with approximately 10 gradient steps per frame.
2. A *dispersion-constrained parameterisation* that reduces the free parameter count from $5N$ to $4N$ and acts as a physics-based regulariser against overfitting, quantified through ablation.
3. An *incidence-angle-weighted observation model* that accounts for the heteroscedastic noise structure of LiDAR measurements on ocean surfaces.
4. An *adaptive model complexity* mechanism that automatically selects the number of spectral components from N=1 for simple swells to N=21 for complex multi-component seas.

Key results. On a physics-based simulation dataset of 100 episodes spanning Beaufort 2–5 sea states (1–30 wave components), WaveSplat achieves:

- **30.1% lower normalised RMSE** than a hyperparameter-tuned Ensemble Transform Kalman Filter (ETKF) baseline (0.223 vs. 0.319, Wilcoxon p=1.1×10^{-15}, Cohen's d=1.16);
- **approximately** 10× **faster** inference (13.4 vs. 132.6 ms/frame, single CPU thread);
- the **only positive Skill Score** in a pre-registered six-metric evaluation panel against a zero-predictor reference, exposing both the ETKF (shape-incoherent) and sliding-window GPR (near-zero predictor) as failing this basic test.

The improvement is highly significant across 100 paired runs with five random seeds, and is robust to both diffuse and specular LiDAR observation models and to extreme observation sparsity (<8% degradation at 20% observation density).

Relevance to Ocean Engineering. This work directly addresses the operational challenge of real-time wave field reconstruction for vessel-scale maritime applications. The method requires no pre-trained models, no offline calibration beyond water depth, and runs on standard CPU hardware at 10-Hz LiDAR frame rates. Results include a companion application to wave-preview feedforward motion control of hydrofoil vessels, demonstrating the practical value of the spectral representation for downstream maritime control systems.

Originality statement. This work is original and is not under consideration elsewhere. The simulation code, all experiment scripts, and five representative episodes are openly available at the project repository to ensure end-to-end reproducibility of all reported tables and figures.

Suggested reviewers. We suggest the following experts in ocean wave sensing and data assimilation (the authors have no conflict of interest with any of them):

- Prof. Tianhao Liu (wave sensing from ship-mounted sensors)
- Prof. Johannes Gemmrich (ocean surface wave observations)
- Dr. Nicolas Desmars (LiDAR-based wave field reconstruction)

We look forward to your consideration of this manuscript.

Sincerely,

Saeyong Park